Lianjie (Marc) Shi

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EDUCATION & QUALIFICATIONS

University of British Columbia (UBC) **Vancouver, Canada**
*Bachelor of Arts, Double Major in Statistics and Psychology (**Cumulative GPA: 92%; WES: 3.9/4.0**)* *09/2020 – 05/2026*

- **Relevant Courses:**

Adv Statistical Inference I (STAT460/560)	95% (A+)	Adv Statistical Inference II (STAT461/561)	94% (A+)
Bayesian Statistics (STAT405)	93% (A+)	Design and Analysis of Experiments (STAT404)	94% (A+)
Time Series & Forecasting (STAT443)	91% (A+)	Real Analysis I (MATH320)	82% (A-)
Calculus IV (MATH317)	93% (A+)	Stochastic Processes (MATH303)	96% (A+)
Linear Regression (STAT306)	90% (A+)	Intro Statistical Inference (STAT305)	98% (A+)
Intro Probability (STAT302)	99% (A+)	Matrix Algebra (MATH221)	93% (A+)
Differential Equations (MATH215)	100% (A+)	Differential Calculus (MATH104)	100% (A+)

- **Honours & Awards:**
  - Dean's List (2023, 2024) – at least 27 credits with grades above 85%;
  - Work Learn International Undergraduate Research Award (WLIURA) – **$8,700** research award (2024);
  - Faculty of Arts International Student Scholarship – **$5,500** merit award (2024);
  - Dr. John and Barbara Petkau Scholarship – **$5,650** merit award for the fourth-year stats student with the highest standing (2025).

Peking University (PKU) **Beijing, China**
Visiting Student (Non-degree) *09/2026 – 01/2027*

- **Relevant Courses (in progress):** Deep Learning and AI; Machine Learning and AI

PUBLICATIONS & CONFERENCES

- Ou, C., Lou, N. M., Maheshka, C., **Shi, M**., Takemura, K., Cheung, B., & Heine, S. J. (2025). *Healthy sleep durations appear to vary across cultures. Proceedings of the National Academy of Sciences*, 122(19), e2419269122. https://doi.org/10.1073/pnas.2419269122
- Zhang, X., Wu, H., **Shi, L.**, & Savalei, V. (in revision). *Model identification made clear: From conceptual foundations to CFA identification rules.* Revise and resubmit at *Psychological Methods*. Preprint: https://osf.io/preprints/psyarxiv/pqa6y
- Zhang, X., Wu, H., Savalei, V., & **Shi, L.** (2026, July). *Model identification in CFA* [Symposium presentation]. In International Meeting of the Psychometric Society (IMPS), Seoul, South Korea.

RESEARCH EXPERIENCE

Directed Studies Project – Cross-Cultural Variation in Healthy Sleep Duration and Chronotype: A Multilevel Analysis
Supervisor: Dr. Steven Heine (Culture and Self Lab, UBC) *09/2022 – 05/2024*

- Fit **mixed-effects (multilevel) models** in R to large-scale cross-national survey data, separating within- and between-country effects of chronotype and sleep duration on health outcomes.
- Estimated **cross-level interactions** between country-level sleep norms and individual sleep duration in predicting health.
- Wrote up the directed-studies project [paper]; contributed as the **4th author** on the published article in *PNAS* (2025).

Research Project – Identifiability and Nested-Model Comparison in Latent Variable Models
Supervisor: Dr. Xijuan (Cathy) Zhang (Quantitative Psychology Lab, York University) *09/2023 – 05/2026*

- Compared competing confirmatory factor models for balanced Likert-scale data, testing **nesting and equivalence relations** between models in R and assessing model fit on empirical data [paper].
- Evaluated **identification rules** for factor models with correlated errors and cross-loadings, applying **Wald rank tests** to check identifiability on simulated and empirical data.
- **Third author** on the resulting methods paper (revise and resubmit, *Psychological Methods*); co-author on the IMPS 2026 talk.

Research Project – Sensitivity of $RMSEA_D$ to Baseline-Model Misspecification in Multi-Group Factor Models
Supervisor: Dr. Victoria Savalei (SEM Lab, UBC) *05/2024 – 08/2025*

- Studied how **baseline-model misspecification** affects tests of **cross-group parameter equality** (measurement invariance) in multi-group factor models, using $RMSEA_D$ (a fit index for nested-model comparison) as the outcome.
- Developed an **R Shiny application** to visualize $RMSEA_D$ across **one- and two-factor models** under user-specified misspecifications (residual covariances, cross-loadings), supporting model comparison and residual analysis.

TECHNICAL PROFICIENCIES

- **R** (Mixed-Effects (Hierarchical) Models, Latent Variable Models, Simulation Studies, R Shiny App Development)
- **Python** (NumPy, pandas, scikit-learn, Matplotlib, PyTorch, TensorFlow/Keras)
- **Java** (Object-Oriented Programming, Data Structures & Algorithms); **C** (Systems Programming, Linux)
- **SQL** (MySQL; Relational Database Design)